Enabling Independent Play through Partial Automation: Designing Software Copilots for Players with Motor Disabilities

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Abstract

Players with upper-limb impairments often face significant barriers when accessing video games, as many games require multiple simultaneous inputs and rapid reactions. One possible solution is partial automation, which delegates inaccessible game actions to a software copilot that automatically executes them. Partial automation, however, has so far been studied only marginally as an accessibility solution. My research therefore investigates how partial automation software copilots can be designed to support players with disabilities during gameplay, focusing on player control, communication, and awareness. The project follows a user-centered methodology, involving players with disabilities to understand their needs and preferences and evaluate proposed solutions. To date, this work has produced initial design recommendations and a research probe for studying partial automation in existing video games.

CCS Concepts

• **Applied computing** → *Computer games*; • **Human-centered computing** → **Accessibility technologies**; **Collaborative interaction**; • **Social and professional topics** → *People with disabilities*.

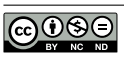
Keywords

Shared control; partial automation; accessible gaming.

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1 Introduction

Video games are an important medium for entertainment, social participation, and self-expression [13, 35, 39]. However, many games remain inaccessible to players with upper-limb impairments because they often impose tight reaction times and require players to coordinate multiple simultaneous inputs [27, 47]. My research investigates partial automation as a way to support these players by delegating selected game actions to a software copilot [3, 11], while preserving player agency and enabling more independent play. The central question is not only whether a software agent can execute game actions, but how such an agent should interact with the player during gameplay. This work therefore studies how software copilots should support player control, communication, and awareness during partial automation gameplay.



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2 Related Work

To access video games, players with upper-limb impairments can adopt various strategies, including control remapping [22, 25], alternative input modalities [5, 26, 34, 46], and accessible input devices [9, 14, 16, 17, 23, 30, 37]. These solutions can make individual inputs easier to perform. However, they do not necessarily reduce the number of actions that the user must control, nor do they relax the timing demands imposed by the game.

When these demands remain inaccessible, players may resort to shared control, delegating selected game actions to a copilot. In shared control, game actions are distributed between a pilot, the primary player, and a copilot, who assists them. When the copilot is human, shared control is referred to as human cooperation; when it is a software agent, as partial automation [3]. Human cooperation is supported by major gaming platforms [29, 41] and used by players with disabilities to access video games [3]. However, it makes play dependent on another person's availability and support, reducing player autonomy. Partial automation, instead, can reduce this dependency, but is available only in selected commercial video games and usually targets specific game tasks [2, 15, 32, 33, 43]. For example, in *Bayonetta* [32, 33], partial automation allows players to delegate character movement, focusing only on attacking. In research, prior work on partial automation has investigated software agents that take over selected game actions to simplify interaction, support players with motor impairments, or dynamically adapt the level of assistance [10, 11, 20, 28]. These systems show that automation can reduce input demands, but they remain limited to specific games or tasks, and may introduce automation confusion, where players struggle to understand which actions they control and which are automated [12].

2.1 Research Gap and Research Questions

In video games, partial automation has so far been mostly studied as a game balancing technique [8, 10, 18, 20, 38, 45], where the goal is to adapt the challenge of the game rather than to address inaccessible input demands. Some studies have considered it as a means to support players with disabilities [11, 28], but they did not

explicitly connect it to human cooperation. This connection, however, is important as both approaches address similar barriers by delegating selected game actions to a copilot [3]. Since human cooperation is already used by players with disabilities, studying how pilots and copilots coordinate during human cooperation gameplay can inform the design of partial automation systems. In this sense, partial automation could preserve the accessibility function of human cooperation while reducing its dependence on a constantly available human copilot. My research therefore **investigates how a software copilot should be designed to provide support comparable to human cooperation, enabling players with disabilities to play independently**. This overarching question is addressed through three research questions:

- **RQ1**: How should the software copilot intervene in gameplay to support players in controlling game actions?
- **RQ2**: How can players and software copilots communicate during partial automation gameplay?
- **RQ3**: How should software copilots support player awareness and prevent automation confusion during partial automation gameplay?

3 Research Approach and Methods

My research is articulated in multiple phases.

3.1 Grounding the Research in Human Cooperation Practices

As general-purpose partial automation systems for game accessibility are not yet available, there are no established practices of players interacting with software copilots. In contrast, human cooperation is already available and relies on the same principle of delegating selected game actions to a copilot. The first formative phase of the research therefore grounded the project in human cooperation through interviews and focus groups with players with disabilities who use this modality, aiming to understand their practices and inform future partial automation systems. This phase has already been completed. The results were analyzed through reflexive thematic analysis [42] and led to the development of a first partial automation system, which serves as a research probe for investigating the three research questions.

3.2 RQ1: Supporting the Control of Game Actions

Designing the copilot's intervention during gameplay involves two main challenges. First, the copilot must decide when and how to intervene, accounting for both the game state and the player's intent, since the same situation may require different support depending on the player's goal. This requires investigating how player intent can be inferred during gameplay and combined with game-state analysis to guide intervention strategies. Second, when the player and the copilot can control the same game actions, their inputs may overlap or conflict and need to be combined through a merging policy. Such a policy may account for factors such as action risk or the difficulties shown by the player during gameplay. These aspects will be investigated through iterative prototyping and empirical studies, comparing how different intervention and merging strategies affect player control, perceived agency, and gameplay experience.

3.3 RQ2: Supporting Communication Between the Pilot and the Copilot

For **RQ2**, the first step will be to understand how players would like to communicate with the copilot and for what purposes. This will include investigating whether they want to ask for information about the game state, express intentions, issue commands, clarify the copilot's behavior, or modify the level of support provided by the system. This preliminary phase will rely on interviews, co-design sessions, and Wizard-of-Oz studies [24], in which players interact with a simulated software copilot while playing. The resulting data will inform the design of communication mechanisms based on natural language interaction, supported by technologies such as large language models.

3.4 RQ3: Supporting Player Awareness

Supporting player awareness requires helping players understand both relevant aspects of the game and the software copilot's behavior. This includes what is happening in the game, which elements require attention, what the copilot is doing, and how its interventions may affect gameplay. Because video games are dynamic and often time-constrained, non-verbal signals, such as visual, auditory, or haptic feedback, can provide faster than explicit verbal explanations. These forms of feedback are already common in video games [6, 7, 31], making them a natural design space to explore. Preliminary elicitation [44] and prototyping studies will therefore be used to understand what information should be communicated, when, and through which modalities. At the same time, these mechanisms will need to remain customizable, so that players can adapt them to their abilities, preferences, and the game-specific demands.

3.5 Evaluation

In all cases, the proposed solutions will be compared with human cooperation, which will serve as a baseline to evaluate how the gameplay experience changes between the two modalities and how effective partial automation support is. The comparison will combine quantitative and qualitative methods. Quantitative measures will include standardized questionnaires, such as NASA-TLX [19] for perceived workload and PXI [1] for player experience, gameplay metrics (*e.g.*, the number of enemies defeated in the two modalities, in a first-person shooter), and system logs to detect conflicts between pilot and copilot actions. Qualitative methods will instead be used to study changes in player behavior, collaboration patterns, perceived control, and coordination between pilot and copilot.

4 Results to Date

To date, the first formative phase has been completed, showing how players with disabilities currently play in human cooperation and what they expect from future partial automation systems [3]. Based on these results, I defined 10 design guidelines and recommendations for shared control systems, including partial automation systems. These guidelines indicate how future systems should support different forms of assistance, collaboration between the pilot

and the copilot, and the transition from human cooperation to partial automation.

These findings also informed the design of *GamePals*, a research probe that enables both human cooperation and partial automation in existing video games. *GamePals* was designed to be easily adaptable to investigate different aspects of partial automation. Indeed, it allows experimenting with different configurations of involved players, subdivisions of game actions, and action merging policies, depending on the specific needs of the research [4]. The system was evaluated with 13 people with upper-limb impairments on two video games – *Rocket League* [36] and *Doom* [21] – to study how the gameplay experience varied between human cooperation and partial automation. The results showed similar coordination patterns across modalities, suggesting that partial automation can reproduce the interaction dynamics observed in human cooperation. At the same time, they also highlighted the importance of communication with the copilot in human cooperation, which was not available in this first implementation of partial automation, as well as some difficulties players experienced when coordinating with the software copilot.

For this reason, I am currently investigating how to improve player awareness during gameplay (RQ3). This includes feedback cues to signal which actions the software copilot intends to perform, as well as signals highlighting key game elements that the player should attend to in order to progress. At the same time, I am studying how to support communication between the pilot and the software copilot (RQ2), allowing players to ask questions about the game and issue commands to the copilot. In the near future, I plan to focus on how the copilot can account for player's actions and difficulties when determining its own behavior and action merging strategies.

5 Open Challenges and Future Research

While my research focuses on designing and evaluating software copilots for partial automation, two broader challenges remain open for future research First, *GamePals* was designed as a research probe: its use requires supervisor support, and partial automation depends on game-specific adaptations. This limits its applicability as a general accessibility solution. Future research should investigate how partial automation can become usable independently by players with disabilities across existing games. This would also help study how different game genres influence pilot–copilot interactions. One possible approach could build on Google's SIMA project [40], which used computer vision and natural language instructions to develop an agent capable of interacting with different 3D video games. Second, my research plan lacks longitudinal studies of how pilot–copilot interaction evolves over time. This limitation is connected to the previous one, as such studies would require a system that players can use independently, over an extended period, and across different games. Longitudinal studies would be important because repeated use may change players' trust and coordination strategies with the copilot, expectations about the support provided, and sense of agency. Understanding these dynamics will be necessary before partial automation can be adopted as a stable accessibility solution.

6 Conclusions

This research investigates partial automation as an accessibility approach for video games, focusing on how software copilots can support player control, communication, and awareness. The project is grounded in human cooperation practices, using *GamePals* as a research probe to design and evaluate software copilots with players with upper-limb impairments. The next stages of the research will refine intervention, communication, and awareness mechanisms through iterative prototyping and empirical studies. Overall, this work aims to inform the design of partial automation systems that preserve player agency while enabling more independent access to video games.

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